

I. Carbohydrates

A. MCQ Carbohydrates Mock Exam February,2012

1. One of the enzymes needed for inter-conversion of sugar in the pentose phosphate pathway is:
 - A. Glucokinase
 - B. Hexokinase
 - C. Transaldolase
 - D. All of the above
2. An important product of pentose phosphate pathway is
 - A. NADH
 - B. NADH₂
 - C. NADPH
 - D. All of the above
3. Absolute configuration of carbohydrates is based upon:
 - A. Hexoses
 - B. D-Glyceraldehyde
 - C. D-ribose
 - D. All
4. Aldose are reducing sugars because in their non-cyclic form contain:
 - A. A ketone group
 - B. An ester group
 - C. An aldehyde group
 - D. A hydroxyl group

5. Kinase reactions:

- A. Inhibit ATP production
- B. Involves the addition of a phosphate group.
- C. Involves the addition of a Ketone group.
- D. Involves the transfer of a hydrogen atom.

B. True and False

- ___6. There are no structural homopolysaccharides.
- ___7. *Carbohydrates can be digested in the stomach.*
- ___8. Fluoride inhibits enolase during glycolysis.

C. Fill in the gap

9. The number of isomers of a monosaccharide can be calculated by the formula _____.

10. An intermediate in triglyceride synthesis is.

- A. Phosphatidic acid
- B. Acyl CoA
- C. Diacyl glycerol
- D. All

11. Sources of NADPH required for de novo fatty acid synthesis include all of the following except:

- A. Pentose phosphate pathway
- B. Malic enzyme
- C. Isocitrate dehydrogenase reaction
- D. Glycolysis

___ 12. Malonyl co A activates fatty acid oxidation.

13. A degradation products of odd numbered fatty acid is:

_____.

14. The immediate precursor of eicosanoids found in the membranes phospholipids is: _____.

15. One method by which amino acid is synthesized:

- A. Transamination
- B. Animonia fixation
- C. Conversion of non essential aminoacid
- D. All the above

C. Fill in the gap

16. Give two examples of aminoacid that can be synthesized from an essential aminoacids

17. The Coenzyme needed during decarboxylation of pyruvate

Is:

- A. Pyridoxyl pyrophosphate
- B. Thiamine pyrophosphate
- C. Vitamin K
- D. All the above

18. ds -DNA in human chromosome is folded _____ times

- A. 2 B. 6 C. 10^4 D. 10^6

19. Bases absorb Light at which wavelength:

- A. Infra red B. Yellow light C. 260nm
- D. All the above

20. Bond that generates primary structure of DNA

- A. N-glycosidic bond
- B. Phosphodiester bond
- C. Van der Waals bond
- D. All the above

21. NDP reductase has _____ binding sites

- A. 1 B. 2 C. 3 D. All the above

22. End product of pyrimidine catabolism is:

- A. Uric Acid B. β - Uredopropione
C. β -Alanine D. All the above

23. One of the following RNA is mostly double stranded

- A. m-RNA B. ribosomal RNA
C. Both of them D. None of them

24. What are ribosomes

_____.

25. Define Translation

_____.

26. Define codon

_____.

27. The percentage of dry weight of protein in human body is:

- A. 10% B. 15% C. 50% D. None

28. List three functional classes of proteins:

29. Symbol of α -helix is

- A. Zig Zag line B. Cylinder
B. Spring D. B & C E. All the above

B. True/False

30. _____ All bonds of tertiary structure exist in
quaternary structure except S-S bond

31. Who are the organs that supply proteolytic enzymes to
gastrointestinal cavity

32.The most important activator of pepsinogen is:

- A.HCl
- B. Pepsin
- C. A & B
- D.None of the above

B. True /False

33. ____The hormone that stimulates the release of pepsinogen is gastrin.

34. ____ L-Amino acids are absorbed by passive diffusion

35.____ γ -glutamate cycle is energy independent transporter.

36.Energy during glutamate cycle is supplied by

- A. Na^+/K^+ ATPase
- B. Glutathione lysis
- C. Glycogen
- D.All the above

37.The glucose monomer in cotton is

- A. α -D
- B. β -D
- C. Both
- D. None

PART VI Vitamins & Coenzymes

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38.A. Coenzyme needed for decarboxylation of aminoacids is

- A. biotin
- B. pyridoxyl pyrophosphate
- C. folic acid
- D. ascorbic acid

39. A vitamin that can be synthesized by symbiotic

bacteria:

- A. Vitamin A
- B. Vitamin D
- C. Vitamin K
- D. All of the above

C.Fill in the gap

40.List three Lipids that can be synthesized from Acetyl CoA:

B.True/False

41.What is the most critical reaction that occur in the human eye:

- A. Conversion of 11-cis retinal to all- trans retinal
- B. 11 -trans retinal to 11-cis retinal
- C. Both the above
- D. None of the above

42.What is the most important function of calcitriol

- A. Absorption of phosphate
- B. Intestinal absorption of calcium
- C. Renal absorption of K^+
- D. All of the above

43.List essential fatty acids:

- _____
- _____

44.What are the importance of ubiquinone.

- A. Reducing agent
- B. Participates in ETS
- C. Inhibits induction of protein synthesis
- D. All of the above

B. True/False

45. _____ Pantothenic acid participates in biosynthesis of acylcarrier protein .

46. Vitamin that supports carboxylation in cytosolic solution is:

- A. Vitamin K
- B. Vitamin C
- C. biotin
- D. All of the above

C. Fill in the gap

47. One of the followings is the function of Vitamin C

- A. Reduction of metallic ion
- B. It is an antioxidant
- C. Participation in hydroxylation reaction
- D. All of the above

48. One of the following enzymes requires lipoic acid

- A. Pyruvate dehydrogenase complex
- B. α -Ketoglutarate dehydrogenase complex
- C. Branched chain amino-acid dehydrogenase complex
- D. All of the above

49.The metal ion which exists in hemoglobin is:

- A.Copper
- B.Iron
- C.Magnesium
- D.All of the above

B. True/False

50._____ Catalyst is any biological molecule that catalyzes chemical reactions.

51. _____ An Oxidase produces H_2O_2 as a by product.

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C. Fill in the gap

52. Classify enzymes based on the number of reaction at a given point:

53. One of the following is a monomeric enzyme:

- A. NDP reductase
- B. Trypsin
- C. Glutamate Synthase
- D. All the above

B. True/False

54. _____ Conformation of an enzyme in the presence and absence of an inhibitor is identical.

55. One of the following is a multifunctional enzyme:

- A. DNA Polymerase I
- B. Fatty acid Synthase
- C. Phosphofructose Kinase 2
- D. All of the above

56. What is an isoenzyme?

True or False

57. List the properties of the enzymes:

58.State the equation that expresses Gibb's free energy

59.During coupled reaction, there is _____

- A.Net release of entropy
- B.Net release of enthalpy
- C.Net release of free energy
- D.None of the above

60.pH optimum for pepsin is

- A.5.0
- B.2.0
- C.9.0
- D.None of the above